# Project Proposal: Pedestrian Tracking

**Task**

Implement a vision system capable of tracking multiple pedestrians from one stationary camera. Evaluate the accuracy (how long can it track a person) on an existing dataset. Number of datasets for this task exist.

**Plan**

- Find a pretrained neural network (YOLOv8) and fine-tune it for **pedestrian detection**
- Implement a multi-object **tracking** algorithm (DeepSORT / ByteTrack).
- **Visualize** trajectories using bounding boxes with IDs and compare them to already existing pedestrian detectors and trackers
- **Tools**: PyTorch, OpenCV, NumPy.
- **Team roles**:
  - **Member 1:** Prepares datasets and fine-tunes the detector.
  - **Member 2:** Implements the tracker and integrates it with the detector.
  - **Member 3:** Evaluates the system, computes metrics, and creates visualizations.

**Evaluation**

The system will be evaluated on pedestrian tracking benchmarks such as MOT17 or MOT20. Standard multi-object tracking metrics (MOTA, IDF1, ID switches, track length) will be used. The proposed method will be compared against a baseline implementation.